

Space-Filling Curves

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1 Main results

Theorem 1.1 (Space-Filling Curve Theorem). For every $m \in \mathbb{N}$ there exists a continuous map $f : [0, 1] \rightarrow [0, 1]^m$ that is surjective.

We will concentrate on result for $m = 2$. For $n \in \mathbb{N}$ we set

$$P_n := \left\{0, \frac{1}{4^n}, \dots, \frac{4^n - 1}{4^n}\right\}$$

and for $k \in P_n$ we set $I_k^n := [k, k + \frac{1}{4^n}]$. Similarly,

$$Q_n := \left\{0, \frac{1}{2^n}, \dots, \frac{2^n - 1}{2^n}\right\} \times \left\{0, \frac{1}{2^n}, \dots, \frac{2^n - 1}{2^n}\right\}$$

and for $(k, l) \in Q_n$ we set $W_{(k,l)}^n := [k, k + \frac{1}{2^n}] \times [l, l + \frac{1}{2^n}]$.

Lemma 1.2. For $n = 0, 1, 2, \dots$ we can find a bijection $\alpha_n : P_n \rightarrow Q_n$ and a continuous map $f_n : [0, 1] \rightarrow [0, 1] \times [0, 1]$ such that the following conditions are satisfied:

1. For each $n \in \mathbb{N}_0$ and each $k \in P_n$ we have $f_n(I_k^n) \subset W_{\alpha_n(k)}^n$.
2. For each $n \in \mathbb{N}_0$ and each $k \in P_n$ the two endpoints of the interval I_k^n get sent to adjacent vertices of the square $W_{\alpha_n(k)}^n$.
3. For each $n \geq m \in \mathbb{N}$ and each $k \in P_n$ and $l \in P_m$ with $I_k^n \subset I_l^m$ we have the inclusion $f_n(I_k^n) \subset W_{\alpha_m(l)}^m$.

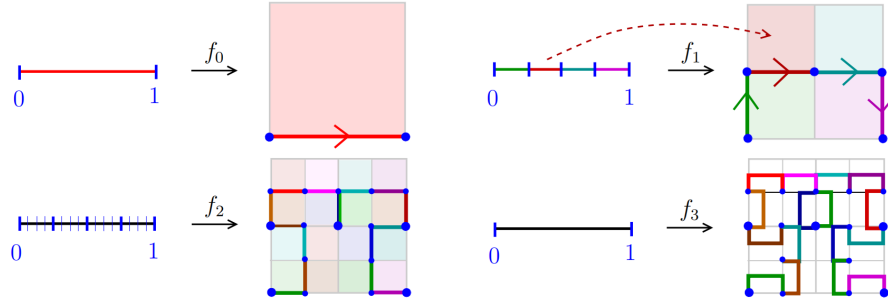


Figure 1: Implicit Maps

Proof. We defined the maps iteratively. For $n = 0$ we have $P_0 = \{0\}$ and $Q_0 = \{(0, 0)\}$, thus we let $\alpha_0 : P_0 \rightarrow Q_0$ be the unique bijection. Furthermore we set

$$\begin{aligned} f_0 : I_0^0 = [0, 1] &\rightarrow W_{(0,0)}^0 = [0, 1] \times [0, 1] \\ t &\mapsto (t, 0) \end{aligned}$$

It is clear that conditions 1 and 2 are satisfied. Now suppose that we have constructed bijections $\alpha_0, \dots, \alpha_n$ and continuous maps $f_0, \dots, f_n$ with the desired properties. Next we need to construct the bijection $\alpha_{n+1} : P_{n+1} \rightarrow Q_{n+1}$ and the maps $f_{n+1} : [0, 1] \rightarrow [0, 1] \times [0, 1]$.

Our first goal is, given $j \in P_n$, to define f_{n+1} on $I_j^n = [k, j + \frac{1}{4^n}]$ and to define α_{n+1} on $j, j + \frac{1}{4^{n+1}}, j + \frac{2}{4^{n+1}}, j + \frac{3}{4^{n+1}} \in P_{n+1}$. We write $\alpha_n(j) =: (k, l) \in Q_n$. It follows easily from 2 that we can find a bijection

$$\begin{aligned} \alpha_{n+1} : & \left\{ j, j + \frac{1}{4^{n+1}}, j + \frac{2}{4^{n+1}}, j + \frac{3}{4^{n+1}} \right\} \\ \rightarrow & \left\{ (k, l), (k + \frac{1}{2^{n+1}}, l), (k, l + \frac{1}{2^{n+1}}), (k + \frac{1}{2^{n+1}}, l + \frac{1}{2^{n+1}}) \right\} \end{aligned}$$

and a map $f_{n+1} : I_j^n \rightarrow W_{(k,l)}^n$ with the following three properties:

1. The map $f_{n+1} : I_j^n \rightarrow W_{(k,l)}^n$ agrees with the map f_n on the endpoints.
2. Each interval $I_{j+\frac{i}{4^{n+1}}}^{n+1}$ gets sent to the square $W_{\alpha_{n+1}(j+\frac{i}{4^{n+1}})}^{n+1}$.
3. The endpoints of each interval $I_{j+\frac{i}{4^{n+1}}}^{n+1}$ get sent to adjacent vertices of the square $W_{\alpha_{n+1}(j+\frac{i}{4^{n+1}})}^{n+1}$.

This can be done in a systematic way as shown in the figure below. These maps

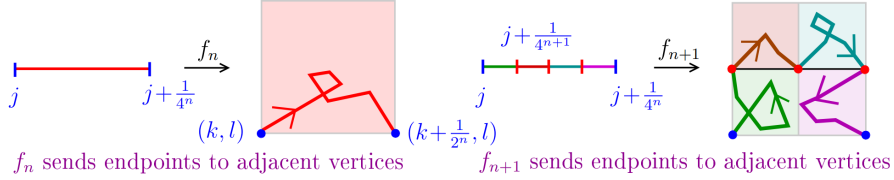


Figure 2: Implicit Maps

$f_{n+1} : I_j^n \rightarrow [0, 1] \times [0, 1]$ agree on the endpoints. Therefore it follows by Lemma A.5 that they define a continuous map $f_{n+1} : [0, 1] \rightarrow [0, 1] \times [0, 1]$. Furthermore the bijections α_{n+1} combined give a bijection $\alpha_{n+1} : P_{n+1} \rightarrow Q_{n+1}$. $\square$

We consider the set $C([0, 1], [0, 1] \times [0, 1])$ of continuous maps from $[0, 1]$ to $[0, 1] \times [0, 1]$. We view this as a metric space with respect to the metric

$$d(f, g) := \sup\{\|f(t) - g(t)\|_{\ell^2} \mid t \in [0, 1]\}.$$

which is well-defined by Proposition A.6.

Lemma 1.3. The sequence $(f_n)_{n \in \mathbb{N}_0}$ of maps $[0, 1] \rightarrow [0, 1] \times [0, 1]$ is a Cauchy sequence in the metric space $C([0, 1], [0, 1] \times [0, 1])$.

Proof. Let $\varepsilon > 0$. We pick $N \in \mathbb{N}_0$ with $\frac{1}{2^N} < \frac{\varepsilon}{2}$. Let $m, n \geq N$. For any $t \in [0, 1]$ we have

$$\begin{aligned} \|f_m(t) - f_n(t)\|_{\ell^2} &\leq \text{diameter of square of side length } \frac{1}{2^N} \\ &= \sqrt{2} \cdot \frac{1}{2^N} < \frac{\varepsilon}{\sqrt{2}} \end{aligned}$$

The first equality follows from property 3 that for $m, n \geq N$ and $t \in [0, 1]$, $f_m(t), f_n(t)$ are contained in a square of length $\frac{1}{2^N}$. $\square$

Corollary 1.4. It follows that

$$\begin{aligned} f : [0, 1] &\rightarrow [0, 1] \times [0, 1] \\ t &\mapsto f(t) = \lim_{n \rightarrow \infty} f_n(t) \end{aligned}$$

is continuous.

Proof. This follows immediately by Proposition A.6. $\square$

Lemma 1.5. The image of the map $f : [0, 1] \rightarrow [0, 1] \times [0, 1]$ is dense.

Proof. Let U be an open subset of $[0, 1] \times [0, 1]$. We need to show that $f([0, 1]) \cap U \neq \emptyset$. It follows easily from an elementary argument, that there exists an $N \in \mathbb{N}_0$ and a $(k, l) \in Q_N$ such that $W_{(k, l)}^N \subset U$. By property 1, there exists an $x \in [0, 1]$ with $f_N(x) \in W_{(k, l)}^N$. By property 1 we know that for each $n \geq N$ we have $f_n(x) \in W_{(k, l)}^N$. Since $W_{(k, l)}^N$ is closed, we see that $f(x) = \lim_{n \rightarrow \infty} f_n(x) \in W_{(k, l)}^N \subset U$. We have thus shown that $f([0, 1]) \cap U \neq \emptyset$. $\square$

Lemma 1.6. The map $f : [0, 1] \rightarrow [0, 1] \times [0, 1]$ is surjective.

Proof. Since $[0, 1]$ is compact and since $[0, 1] \times [0, 1]$ is Hausdorff we obtain from Proposition A.2 that $f([0, 1])$ is a closed subset of $[0, 1] \times [0, 1]$. By Lemma above, we know that $f([0, 1])$ is dense in $[0, 1] \times [0, 1]$. Thus, it follows immediately from the definition of a dense subset that $f([0, 1]) = [0, 1] \times [0, 1]$. $\square$

Proof of Theorem 1.1 for $m \geq 2$. We need to show that for each $n \geq 2$, there exists a surjective continuous map $[0, 1] \rightarrow [0, 1]^m$. We prove this by induction on m . By the above we know that there exists a surjective continuous map $f : [0, 1] \rightarrow [0, 1]^2$. Now suppose we have found a surjective continuous map $\varphi : [0, 1] \rightarrow [0, 1]^m$ for some $m \geq 2$. The map

$$[0, 1] \xrightarrow{f} [0, 1] \times [0, 1] \xrightarrow{\text{id} \times \varphi} [0, 1] \times [0, 1]^m = [0, 1]^{m+1}$$

is evidently surjective. $\square$

Remark. Note that f is nowhere differentiable but continuous everywhere!

Corollary 1.7. For every $n \in \mathbb{N}$ there exists a surjective continuous map $\mathbb{R} \rightarrow \mathbb{R}^n$.

Remark. Let $m \geq 2$, we know that $[0, 1]$ and $[0, 1]^m$ are not homeomorphic so we could not find any bijection between the two maps (since bijection between two normal sets must be a homeomorphism). Whereas, we do have some similar results compared to Space-Filling Curve Theorem.

Theorem 1.8. There exists an injective continuous map $f : [0, 1] \rightarrow [0, 1]^2$ such that $f([0, 1])$ has non-zero 2-dimensional Lebesgue measure.

A Some results in Analysis and Topology

Lemma A.1. Let X be a topological space then

1. If X is compact, then any closed subset of X is also compact.
2. Let $A \subset X$ be a compact subset. If X is Hausdorff, then A is a closed subset of X .

Proof.

1. Let X be a compact space and let A be a closed subset of X . Let $\{U_i\}_{i \in I}$ be an open cover of A . By definition of the subset topology there exists for every $i \in I$ an open subset V_i of X such that $U_i = V_i \cap A$. Since A is closed then $X \setminus A$ is open. The open subset $\{V_i\}_{i \in I}$ together with $X \setminus A$ are open cover of X . Since X is compact, we can cover X by finitely many of these open subsets. But then we can also cover A by finitely many of the U_i .
2. Let X be a Hausdorff space and let $A \subset X$ be a compact subset. We want to show that $X \setminus A$ is open. It is enough to show that given $x \in X \setminus A$, then there exists an open neighbourhood V of x that is contained in $X \setminus A$. We apply the Hausdorff property to $x \in X \setminus A$ and $y \in A$. For every $y \in A$ we obtain disjoint open neighbourhoods U_y of y and V_y of x . Evidently we have

$$A = \bigcup_{y \in A} \{y\} \subset \bigcup_{y \in A} (U_y \cap A) \subset A$$

Thus, we see that $\{U_y \cap A\}_{y \in A}$ is an open cover of A . Since A is compact then there exists $y_1, \dots, y_k \in A$ such that

$$A = \bigcup_{i=1}^k (U_{y_i} \cap A)$$

Now we consider

$$V := \bigcap_{i=1}^k V_{y_i}$$

Since V is the intersection of finitely many open subsets, it is open itself. In addition, V does not intersect any of the U_{y_i} , $i = 1, \dots, k$ by definition. This indicates that V is disjoint from $A \subset U_{y_1} \cup \dots \cup U_{y_k}$. $\square$

Proposition A.2. Let $f : X \rightarrow Y$ be a continuous map between topological spaces. If X is compact and if Y is Hausdorff, then the map $f : X \rightarrow Y$ is closed.

Proof. Let A be a closed subset of X . We need to show that $f(A)$ is closed subset of Y . In fact we see that from Lemma above

$$A \subset X \text{ closed} \implies A \text{ compact} \implies f(A) \text{ compact} \implies f(A) \text{ closed}$$

$\square$

Lemma A.3. Let X be a topological space and let $\{X_i\}_{i \in I}$ be a family of subsets. The following two statements are equivalent:

1. For every $U \subset X$ we have: U is open if each $U \cap X_i$ is open in X_i .
2. For every $U \subset X$ we have: U is closed if each $U \cap X_i$ is closed in X_i .

in addition if statements 1 or 2 holds, then the following statement holds:

3. A map $f : X \rightarrow Y$ to another topological space Y is continuous if each restriction $f|_{X_i} : X_i \rightarrow Y$ is continuous.

Proof. (1 $\implies$ 2) : We assume that statement 1 holds. Now let $U \subset X$ be a subset, we have

$$\begin{aligned} \forall i \in I, X_i \cap U \text{ is closed in } X_i &\implies \forall i \in I, (X \setminus U) \cap X_i \text{ is open in } X_i \\ &\implies X \setminus U \text{ is open in } X \implies U \text{ is closed in } X \end{aligned}$$

(2 $\implies$ 1) : Basically the same as above.

(1 $\implies$ 3) : Let $f : X \rightarrow Y$ be a map to some topological space Y . We suppose that each $f|_{X_i}$ is continuous. Let $U \subset Y$ be an open subset. We need to show that $f^{-1}(U)$ is open subset of X . By statement 1, it's enough to show that each intersection $f^{-1}(U) \cap X_i$ is an open subset of X_i . But this follows from the fact that $f^{-1}(U) \cap X_i = (f|_{X_i})^{-1}(U)$ and the hypothesis that $f|_{X_i}$ is continuous. $\square$

Lemma A.4. Let X be a topological space and let M be a subset of X .

1. Let $\{U_i\}_{i \in I}$ be an arbitrary open cover of X . The following two statements hold:
 - (a) M is open if and only if each intersection $M \cap U_i$ is open in U_i .
 - (b) M is closed if and only if each intersection $M \cap U_i$ is closed in U_i .
2. Let $\{A_i\}_{i \in I}$ be a locally finite closed cover of X . The following two statements hold:
 - (a) M is open if and only if each intersection $M \cap A_i$ is open in A_i .
 - (b) M is closed if and only if each intersection $M \cap A_i$ is closed in A_i .

Proof. Let X be a topological space and let M be a subset of X . We will only deal with the case for M (open/closed), the statement for (closed/open) subsets follows immediately by the fact $(X \setminus M) \cap U_i = U_i \setminus (M \cap U_i)$.

1. Let $\{U_i\}_{i \in I}$ be an open cover of X .
 - ($\implies$) : Suppose M is open, it follows by definition of subspace topology that each intersection $M \cap U_i$ is open in U_i .
 - ($\impliedby$) : Suppose that for every $i \in I$ the intersection $M \cap U_i$ is an open subset of U_i . By definition of the subspace topology this means that for

each $i \in I$ there exists an open subset Z_i of X with $Z_i \cap U_i = M \cap U_i$. We have

$$M = \bigcup_{i \in I} (M \cap U_i) = \bigcup_{i \in I} (Z_i \cap U_i)$$

since Z_i and U_i are open subsets in X we see that M is an open subset of X .

2. Let $\{A_i\}_{i \in I}$ be a locally finite closed cover of X .
 $(\implies)$ This follows immediately by the definition of subspace topology.
 $(\impliedby)$ Suppose we are given a finite closed cover $\{A_i\}_{i=1, \dots, m}$ of X such that for each i , the intersection $M \cap A_i$ is a closed subset of A_i . It follows that for each $i \in \{1, \dots, m\}$ there exists a closed subset Z_i of X with $Z_i \cap A_i = M \cap A_i$. We have

$$M = \bigcup_{i=1}^m (M \cap A_i) = \bigcup_{i=1}^m (Z_i \cap A_i) = \text{closed subset of } X$$

Now suppose that we are given a locally finite closed cover $\{B_i\}_{i \in I}$ of X such that for each $i \in I$ the intersection $M \cap B_i$ is a closed subset of B_i . Since the cover is locally finite there exists for each $x \in X$ an open neighbourhood W_x such that $I_x := \{i \in I \mid W_x \cap B_i \neq \emptyset\}$ is finite. Since $\{W_x \cap B_i\}_{i \in I_x}$ is a finite closed cover of W_x we obtain from the above that $M \cap W_x$ is closed. But since $\{W_x\}_{x \in X}$ is an open cover of X we obtain from (1b) that M is a closed subset of X . $\square$

Lemma A.5. Let $f : X \rightarrow Y$ be a map between topological spaces.

1. If there exists an open cover $\{W_i\}_{i \in I}$ such that each restriction $f|_{W_i}$ is continuous then f itself is continuous.
2. If there exists a locally finite closed cover $\{A_i\}_{i \in I}$ of X such that each restriction $f|_{A_i}$ is continuous, then f itself is continuous.
3. If X is Hausdorff and if there exists a locally finite cover $\{A_i\}_{i \in I}$ of X such that each A_i is compact and such that each restriction $f|_{A_i}$ is continuous, then f itself is continuous.

Proof. Let $f : X \rightarrow Y$ be a map between topological spaces.

1. This result follows by Lemma A.4 (1a) and Lemma A.3 (1 $\implies$ 3)
2. This follows by Lemma A.4 (2a) and Lemma A.3 (2 $\implies$ 3).
3. This follows from Lemma A.4 and Lemma A.1. $\square$

Proposition A.6. Let (M, d) be a metric space and let X be a non-empty topological space. The map

$$\begin{aligned} \tilde{d} : C(X, M) \times C(X, M) &\rightarrow \mathbb{R}_{\geq 0} \\ (f, g) &\mapsto \tilde{d}(f, g) := \sup\{d(f(x), g(x)) \mid x \in X\} \end{aligned}$$

has the following properties:

1. The map $\tilde{d}$ defines a metric on $C(X, M)$.
2. If X is compact, the the map

$$\begin{aligned}\tilde{d} : C(X, M) \times C(X, M) &\rightarrow \mathbb{R}_{\geq 0} \\ (f, g) &\mapsto \tilde{d}(f, g) := \sup\{d(f(x), g(x)) \mid x \in X\}\end{aligned}$$

is also a metric on $C(X, M)$.

3. If (M, d) is complete, then given any Cauchy sequence $(f_n)_{n \in \mathbb{N}}$ in $(C(X, M), \tilde{d})$ the map

$$\begin{aligned}f : X &\rightarrow M \\ x &\mapsto \lim_{n \rightarrow \infty} f_n(x)\end{aligned}$$

exists, it is a continuous map and in the metric space $(C(X, M), \tilde{d})$ and it is the limit of the sequence $(f_n)_{n \in \mathbb{N}}$.

4. If (M, d) is complete, then so is $(C(X, M), \tilde{d})$.

Proof. The proof for statements 1 and 2 should be very straightforward by definition so we leave this as an exercise for the readers. Statement 4 follows immediately from statement 3, thus it remains to prove the third statement of the proposition. Let $(f_n)_{n \in \mathbb{N}}$ be a Cauchy sequence in $(C(X, M), \tilde{d})$.

Let $x \in X$. Note that we have $d(f_n(x), f_m(x)) \leq \tilde{d}(f_n, f_m)$. Thus we see that $(f_n(x))_{n \in \mathbb{N}}$ is a Cauchy sequence in the metric space (M, d) . By our hypothesis (M, d) is complete, so $(f_n(x))_{n \in \mathbb{N}}$ converges in (M, d) to a point we call $f(x)$. This is true for all $x \in X$ so

$$\tilde{d}(f, f_n) = \sup\{d(f(x), f_n(x)) \mid x \in X\} \rightarrow 0$$

as $n \rightarrow \infty$. Thus, f is the limit of the sequence $(f_n)_{n \in \mathbb{N}}$ in the metric space $(C(X, M), \tilde{d})$.

Now we show that $f : X \rightarrow M$ is continuous. Let $U \subset M$ be an open subset. We need to show that $f^{-1}(U)$ is open. It suffices to show that for every $x \in f^{-1}(U)$ there exists an open neighbourhood W of $x \in X$ with $f(W) \subset U$. Thus let $x \in f^{-1}(U)$. Since $U \subset M$ is open, there exists an $\varepsilon > 0$ such that $B_\varepsilon^\circ(f(x)) \subset U$. By previous results, there exists an $n \in \mathbb{N}$ such that for every $x \in X$ we have $d(f_n(x), f_m(x)) < \frac{\varepsilon}{3}$. Furthermore, by hypothesis $f_n : X \rightarrow M$ is continuous, which implies that there exists an open neighbourhood W of x with $f_n(W) \subset B_{\varepsilon/3}^\circ(f_n(x))$. Now we see that for each $y \in W$ we have

$$d(f(x), f(y)) \leq d(f(x), f_n(x)) + d(f_n(x), f_n(y)) + d(f_n(y), f(y)) < \varepsilon.$$

But this means precisely that $f(W) \subset B_\varepsilon^\circ(f(x)) \subset U$. □